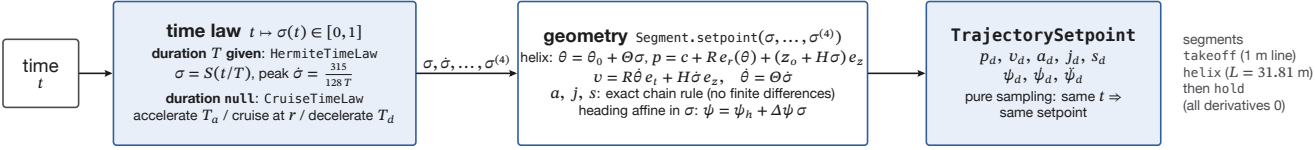
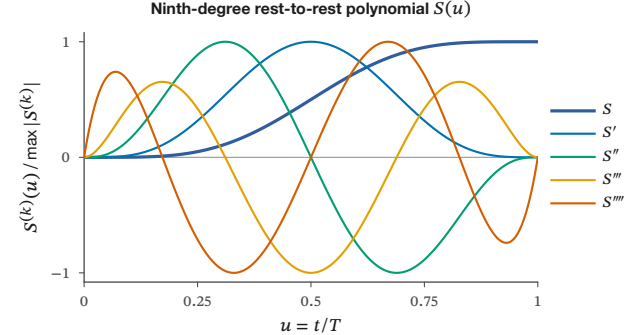


a A segment = fixed geometric path + rest-to-rest time law trajectories/base.py, timing.py, helix.py



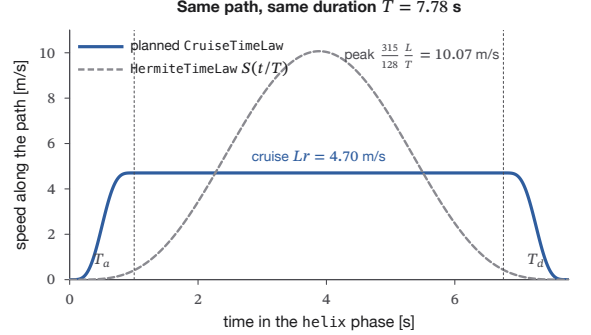
Changing the time law changes only *how fast* the same curve is traversed: $dz/d\theta = H/\Theta$ is fixed, so the helix pitch never depends on timing. Every law starts and ends at rest with four zero derivatives, so joints between segments are C^4 and jerk/snap feedforward stays continuous.

b



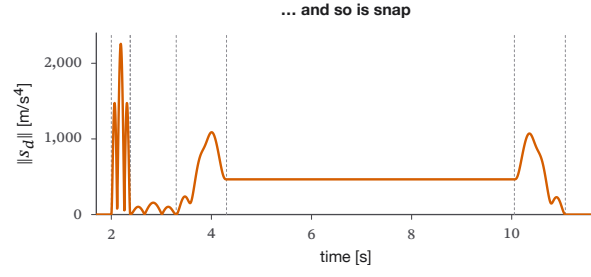
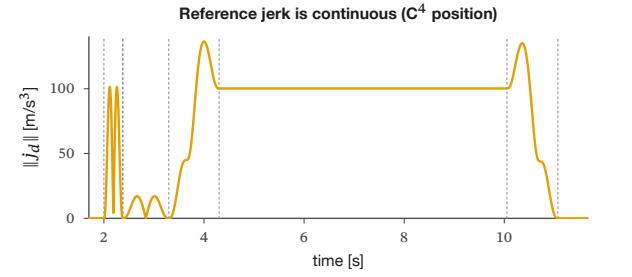
$S(u) = 126u^5 - 420u^6 + 540u^7 - 315u^8 + 70u^9$, $S'(u) = 630u^4(1-u)^4$
 $S^{(k)}(0) = S^{(k)}(1) = 0$ for $k = 1..4$: the unique degree-9 polynomial with these 10 end conditions (Hermite interpolation; not claimed to be minimum-snap). Normalisers: 2.461, 9.37, 78.8, 623.

c



Cruise law: $\dot{\sigma} = r S(t/T_d)$, then r , then $r(1 - S(t'/T_d))$; $T_c = 1/r - (T_d + T_d)/2$. With the time fixed, a Hermite law needs $2.14 \times$ the peak speed $\Rightarrow$ centripetal load $\propto v^2$.

d



Dashed lines: joints between time-law pieces (takeoff accelerate/decelerate, helix accelerate/cruise/decelerate, hold). j_d and s_d enter the geometric controller as $F_d = m j_d + \dots$ and $F_d = m s_d + \dots$, i.e. desired body rate and angular acceleration: a jump here would be a torque step. Jerk and snap peak on the short minimum-time ramps; while cruising at constant speed they are constant, not zero, because the centripetal acceleration rotates: $\|j_d\| = R\dot{\theta}^3, \|s_d\| = R\dot{\theta}^4$ ($\dot{\theta} = 4.645$ rad/s).